Dynamic Reinforcement Learning–Based Handover Control for UAV-Assisted 6G Networks

Project Report Submitted in Partial Fulfilment of the Requirements for the Degree of

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

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April 2026

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Abstract

Sixth-generation (6G) wireless networks demand seamless mobility management under ultra-dense, multi-tier heterogeneous deployments operating across Sub-THz and mmWave bands. While context-aware reinforcement learning approaches such as Linear UCB (LinUCB) have demonstrated strong handover performance in static heterogeneous networks — achieving an 11.1This work extends the LinUCB-based 6G handover framework by introducing a drone base station (Drone-BS) tier in which each aerial node is governed by an independent Policy Gradient (REINFORCE with baseline) agent that learns to autonomously reposition in three-dimensional space. A realistic battery lifecycle model drives return-to-base and recharging cycles, and the drone's state representation explicitly encodes state-of-charge and distance-to-charger features, enabling the policy to internalize energy-coverage trade-offs without manual tuning. The resulting dual-layer architecture — LinUCB for per-step UE association and REINFORCE for drone trajectory optimization — creates a cooperative feedback loop where drone repositioning continuously improves the channel geometry available to the association agent. Over a 3000 × 3000 m² grid with 300 UEs and 6 drone BSs, the system achieves an additional 2–3 dB SINR gain, 10–15

Index Terms — 6G, heterogeneous networks, mobility management, handover optimization, Sub-THz/mmWave, reinforcement learning, contextual multi-armed bandit, LinUCB, beamforming, SINR,Unmanned Aerial Vehicle (UAV), Policy Gradient

I. Introduction

The Handover problem refers to the difficulties that arise when a User Equipment (UE) transitions its connection from one base station (BS) to another while moving across a cellular network. Ideally, this transfer should be seamless, but in practice it often suffers from delays, poor timing, or incorrect cell selection. When the handover occurs too late or the target cell fails to acquire the UE, users experience disruptions such as dropped connections, frozen video playback, reduced data rates, or increased latency. These issues directly affect applications that rely on continuous connectivity, making handover reliability essential to maintaining overall quality of service (QoS).

Sixth-generation (6G) networks pushes these requirements even further. By moving towards sub-THz frequencies, extremely dense deployments, and highly directional beams, 6G aims to support ultra-high data rates, real-time applications, and seamless connectivity in dynamic environments. However, the same characteristics that enable these capabilities also amplify the difficulty of performing reliable handovers. Narrow beams and fast-changing channel conditions make link quality highly sensitive to even small movements or environmental variations. In such settings, traditional mechanisms based on simple threshold comparisons are often inadequate, as they cannot adapt quickly enough to the pace of change or anticipate when a link is likely to deteriorate.

In this context, **Reinforcement Learning (RL)** provides a promising path forward for enhancing handover performance in next-generation networks. Rather than relying on pre-defined rules, an RL system learns from experience, observing how signal conditions evolve and how user movement affects link stability. The handover task can be naturally viewed through the lens of a Contextual Multi-Armed Bandit (CMAB) problem, where each base station represents a possible action and the agent must select the most suitable one based on the current context. By continuously learning from the outcomes of its decisions, the agent becomes increasingly capable of choosing the best cell at the right moment, reducing unnecessary handovers, avoiding ping-pong effects, and maintaining more stable connections even under rapidly changing conditions.

Drone Base Station (Drone-BS) is a fully functional radio access node mounted on an Unmanned Aerial Vehicle (UAV), typically a rotary-wing platform such as a quadrotor, that can hover in place or move through three-dimensional space while actively serving ground users. Unlike conventional terrestrial base stations, which are permanently fixed to towers or rooftops, a drone BS can reposition itself in real time by adjusting its horizontal coordinates and altitude. This mobility gives it a significant advantage in dynamic environments: operating at elevation reduces obstruction by buildings and terrain, increasing the probability of line-of-sight links to ground users and improving the effectiveness of directional beamforming. In 6G networks, where Sub-THz signals are highly susceptible to blockage and coverage radii are small, the ability of a drone BS to physically move toward a user cluster, fill a coverage void, or redistribute load away from a congested cell makes it a valuable complement to static infrastructure.

Policy Gradient methods optimise an agent's behaviour by directly adjusting the parameters of its policy — the function mapping observed states to actions — through gradient ascent on expected cumulative reward. Unlike value-based methods such as Q-learning, they treat the policy as a parameterised probability distribution over actions, making them naturally suited to continuous action spaces where discretisation is impractical. In this work, three policy gradient variants are employed for drone base station mobility optimisation. The foundational approach is REINFORCE with baseline, which collects experience over a short horizon and uses a learned value function to reduce the variance of gradient estimates, producing more stable updates. To address the limitations of Monte Carlo returns in the presence of abrupt battery state transitions, an Advantage Actor-Critic (A2C) variant is also implemented, which updates the policy at every step using a bootstrapped estimate of the advantage rather than waiting for a full trajectory. Finally, a Actor-Critic (SAC) variant is evaluated, which additionally incorporates an entropy bonus to encourage exploration, an off-policy replay buffer to make better use of the limited 200-step simulation horizon, and twin critic networks to stabilise value estimation. All three agents share the same state representation — encoding drone position, network load, coverage quality, and battery state-of-charge — and the same Gaussian policy architecture, differing only in how they compute and apply gradient updates.

II. Literature Review

A. Evolution of Cellular Networks and Handover Challenges

The handover problem has evolved significantly across generations of cellular networks, with each advancement introducing new complexities in mobility management. Traditional handover mechanisms in 4G and early 5G networks primarily relied on fixed signal thresholds and rule-based decision-making, where handover decisions were triggered when signal quality metrics such as Reference Signal Received Power (RSRP) or Signal-to-Interference-plus-Noise Ratio (SINR) crossed predefined thresholds. While these approaches provided adequate performance in relatively stable environments, they exhibit significant limitations in modern wireless networks characterised by high mobility, dense deployments, and rapidly changing channel conditions.

Fifth-generation networks introduced advanced features including millimeter-wave (mmWave) communication, massive MIMO, and dense small cell deployments to achieve higher data rates and improved network efficiency [**?**]. However, these enhancements simultaneously amplified the complexity of mobility management. Rappaport et al. [**?**] demonstrated that the short range and high sensitivity of mmWave signals require more frequent handovers with greater precision, while narrow pencil beams can be easily disrupted by blockages or small changes in user equipment orientation.

The theoretical foundation for sequential decision-making under uncertainty — directly applicable to the handover selection problem — was established by Auer et al. [**?**], who proved finite-time regret bounds for the Upper Confidence Bound (UCB) algorithm. Their work formalised the principle that an agent should balance selecting the empirically best arm (exploitation) with visiting less-tried options to reduce uncertainty (exploration). This principle maps directly onto handover selection, where each candidate base station constitutes a bandit arm with an uncertain reward distribution that must be learned online.

B. 5G Deployment Challenges and Network Densification

The deployment of ultra-dense heterogeneous networks (HetNets) in 5G created new challenges for handover optimisation. Uusitalo et al. [**?**] analysed 5G network coverage planning and identified deployment challenges related to inter-tier handover management, load balancing, and interference coordination across macro, micro, and pico cell layers. The work by Martín-Vega et al. [**?**] developed system models for average downlink SINR in multi-beam 5G networks, revealing that beam-based communication introduces additional complexity in predicting handover timing and target cell selection. Their analysis showed that SINR at cell boundaries is highly sensitive to beam alignment quality, motivating the use of real-time contextual features in handover decision-making.

Large-scale propagation measurements conducted by Rappaport et al. [**?**] provided critical insights into distance-dependent path loss models for outdoor and indoor 5G systems. Their findings indicate that mmWave frequencies experience significantly higher path loss and shadowing effects compared to sub-6 GHz bands, necessitating more intelligent handover strategies that can anticipate channel quality degradation before complete link failure occurs. These empirical models inform the propagation pipeline used in this work, where separate LOS and NLOS path loss expressions with frequency-dependent offsets are applied across the three ground base station tiers.

C. 6G Vision, Sub-THz Communication, and Drone Base Stations

Sixth-generation networks aim to push beyond the capabilities of 5G by targeting sub-THz frequencies (100 GHz to 1 THz) to achieve ultra-high data rates exceeding 100 Gbps, sub-millisecond latency, and massive connectivity [**?**]. Rappaport et al. [**?**] provided comprehensive analysis of sub-THz channel propagation, highlighting that these frequencies experience severe atmospheric absorption, requiring highly directional beamforming with extreme angular precision. Their demonstration that path loss at 142 GHz can exceed that at 28 GHz by more than 20 dB over typical cell ranges motivates the molecular absorption coefficients adopted in this simulation (0.5 dB/km at 95 GHz and 2.0 dB/km at 140 GHz).

The integration of unmanned aerial vehicles (UAVs) as aerial base stations represents a natural response to 6G coverage and capacity demands. Zeng et al. [?] established the foundational framework for UAV-assisted wireless communications, identifying fundamental tradeoffs between UAV altitude, which improves line-of-sight (LOS) probability but increases path length, speed, which affects Doppler spread and energy consumption, and coverage area. Their work demonstrated that an optimal UAV deployment altitude exists that maximises ground coverage subject to a minimum SINR constraint — a principle directly exploited in the altitude correction applied in this simulation, where drone base stations operating above 50 m receive a path loss benefit reflecting improved LOS probability.

Mozaffari et al. [?] provided a comprehensive tutorial on UAV communications, covering deployment strategies for both static and mobile UAV base stations, interference management in UAV-assisted HetNets, and energy efficiency considerations. Their three-dimensional placement optimisation framework forms the conceptual basis for the fourth-tier drone base stations introduced in this work. Critically, their energy model — which accounts separately for propulsion power and RF electronics overhead — directly informed the battery model implemented here, where hover power (800 W), horizontal and vertical movement power, and transmission overhead (15 W) are tracked independently per simulation step.

Al-Hourani et al. [?] derived the optimal low-altitude platform (LAP) altitude for maximum coverage as a function of environment type. Their probabilistic LOS model, which expresses LOS probability as a sigmoidal function of elevation angle and environment parameters, underpins the distance-, frequency-, and base-station-type-dependent LOS probability function implemented in this simulation. Their finding that higher altitude monotonically improves LOS probability — up to the point where coverage area shrinks due to geometric constraints — explains the altitude advantage incorporated into the drone path loss correction above 50 m.

D. Reinforcement Learning for Handover Optimisation

Recognising the limitations of traditional threshold-based approaches, researchers have increasingly explored reinforcement learning (RL) techniques for adaptive handover management. Yajnanarayana et al. [?] pioneered the application of RL to 5G handover control, proposing a centralised agent that processes radio measurement reports and selects handover actions to maximise long-term utility. Their simulation results demonstrated significant improvements in handover success rate and throughput compared to conventional 3GPP handover mechanisms.

Polese et al. [?] provided a comprehensive survey of RL applications in 5G and beyond networks, categorising approaches into model-free methods (Q-learning, SARSA, policy gradient) and model-based techniques. They identified key challenges including sample efficiency, scalability to large state spaces, and the need for safe exploration during the learning phase. The survey emphasised that contextual bandit formulations offer a promising middle ground between full Markov Decision Processes and simple multi-armed bandits, providing sufficient learning capability while maintaining computational tractability.

The theoretical foundations underpinning both the LinUCB and policy gradient components of this work are comprehensively treated by Sutton and Barto [?]. The policy gradient theorem, developed in Chapter 13 of their text, establishes that the gradient of expected return with respect to policy parameters is proportional to the expected product of the log-probability of selected actions and their discounted returns. This result directly justifies the REINFORCE update rule implemented in the drone policy network, where gradients are computed manually through backpropagation over a two-layer MLP with separate policy and value heads.

E. Multi-Armed Bandit Approaches

Multi-armed bandit (MAB) frameworks have emerged as particularly suitable for handover optimisation due to their ability to balance exploration and exploitation in sequential decision-making. Chen et al. [?] applied MAB techniques to mobility management in ultra-dense networks, demonstrating that UCB algorithms can effectively reduce handover frequency while maintaining quality of service. However, their context-free approach does not exploit real-time channel state information, potentially limiting performance in highly dynamic environments.

The foundational algorithm for contextual bandits used in this work — Linear UCB (LinUCB) — was introduced by Li et al. [?], who applied it to personalised news article recommendation. LinUCB assumes a linear relationship between a context feature vector and the expected reward for each arm, and maintains a per-arm ridge regression model that is updated with each observation. The upper confidence bound on the expected reward is then computed as the sum of the estimated reward and an uncertainty term derived from the inverse covariance matrix of the observed contexts. This formulation provides both a theoretically grounded exploration bonus and an efficient closed-form update — properties that make it well suited to real-time handover decisions where low computational overhead is essential.

Sun et al. [?] extended MAB approaches by incorporating spatial and temporal context into handover decisions for ultra-dense mmWave networks. Their Contextual Multi-Armed Bandit (CMAB) formulation uses features such as received signal strength, user location, and cell load to guide base station selection. By formulating two MAB problems focusing on spatial and space-time contexts respectively, their simulation results demonstrated that contextual handover mechanisms significantly outperform context-free counterparts across all evaluated scenarios.

F. Policy Gradient Methods for Continuous Control

The policy gradient method employed for drone mobility optimisation in this work — REINFORCE with a learned value baseline — was originally proposed by Williams [?]. Williams demonstrated that the gradient of expected cumulative reward with respect to policy parameters can be estimated by Monte Carlo rollouts, yielding the fundamental update rule:

$$\nabla_\theta J(\theta) = \mathbb{E}_\pi \left[\nabla_\theta \log \pi_\theta(a_t|s_t) \cdot G_t \right], \tag{1}$$

where $G_t = \sum_{k=0}^{T-t} \gamma^k r_{t+k}$ is the discounted return from step t, π_θ is the stochastic policy parameterised by θ, and the expectation is taken over trajectories generated by π_θ. The principal weakness of vanilla REINFORCE is high variance in the gradient estimates, which slows convergence. This is addressed in the present work through two mechanisms: subtraction of a learned state-value baseline $V_\phi(s_t)$ — reducing each return to an advantage estimate $A_t = G_t - V_\phi(s_t)$ — and normalisation of returns to zero mean and unit variance within each update batch.

The policy itself is represented as a Gaussian distribution over continuous three-dimensional movement actions $(\Delta x, \Delta y, \Delta z)$, with mean $\mu_\theta(s)$ produced by a two-layer MLP and a learnable log-standard deviation $\log \sigma$. This Gaussian policy parameterisation for continuous action spaces is standard in the actor-critic and policy gradient literature [?] and is well suited to the drone repositioning problem, where the action space is inherently continuous and bounded.

G. Deep Reinforcement Learning Approaches

More recent work has explored deep reinforcement learning (DRL) methods that can handle high-dimensional state spaces and complex reward structures. Machumilane et al. [?] developed a DRL-based approach for handover and SINR-aware path optimisation in 5G UAV mmWave communication. Their Deep Q-Network (DQN) agent learns optimal handover policies while simultaneously optimising UAV trajectory to maintain connectivity. Results demonstrated significant improvements in both handover efficiency and end-to-end throughput for mobile aerial platforms. However, DRL approaches face several practical challenges: they require extensive training data, their black-box nature limits interpretability, and they may exhibit instability during online learning. The REINFORCE-with-baseline approach used in this work retains the interpretability and convergence guarantees of simpler linear models while still learning continuous drone movement policies online.

H. Energy-Aware UAV Communications

Battery and energy management are critical operational constraints for drone base stations. Zeng and Zhang [?] addressed energy-efficient UAV communication with trajectory optimisation, deriving the fundamental relationship between UAV propulsion power, speed, and altitude. Their model, which decomposes UAV power consumption into blade profile power, induced power, and parasite power components, established that hover power dominates at low speeds while aerodynamic drag dominates at high speeds — motivating the separate HOVER_POWER_W and MOVE_POWER_COEFF terms in the battery model of this work. Their trajectory optimisation results showed that energy-aware path planning can extend UAV operational time by over 40% compared to naive straight-line trajectories.

The battery-aware state machine implemented in this work — comprising ACTIVE, RTB (Return-To-Base), and CHARGING states with energy-dependent transitions — reflects the operational lifecycle described in [?] and [?]. The reward function for the drone policy explicitly penalises low battery states with an exponential urgency term, encouraging the agent to learn proactive return-to-base behaviour rather than relying on forced transitions at the critical battery threshold. This design aligns with the finding of Zeng and Zhang [?] that proactive energy management yields substantially better coverage continuity than reactive strategies.

I. Research Gaps and Motivation

While existing literature has made significant progress in applying learning-based techniques to handover optimisation, several critical gaps remain, particularly in the context of 6G networks with aerial infrastructure.

Limited 6G-specific modelling with drone BSs: Most existing RL approaches focus on 5G mmWave scenarios with static ground infrastructure and do not address sub-THz propagation characteristics, molecular absorption, or the joint optimisation of ground HetNet handover and drone BS placement simultaneously.

Absence of battery-aware drone RL: Prior UAV-BS works [?], [?] treat energy constraints as a fixed operational parameter rather than integrating battery state directly into the RL reward function and policy state. This work addresses this gap by incorporating battery state-of-charge and distance to charging station as policy features, enabling the drone to learn energy-aware repositioning behaviour.

Context feature engineering for 6G: Previous CMAB approaches [?], [?] have not systematically investigated which contextual features are most informative in four-tier HetNets that include mobile aerial base stations. The LinUCB implementation here operates over a ten-dimensional context vector that includes drone-specific features such as elevation angle and aerial frequency band indicator.

Dual RL system co-optimisation: No prior work has simultaneously deployed a contextual bandit for UE association and a policy gradient agent for drone positioning within a single simulation, leaving open the question of how these two RL systems interact and whether their joint optimisation is stable. This work demonstrates that the two agents co-exist stably, with LinUCB adapting its association decisions as drone positions shift and the drone policy learning to position itself where LinUCB is most likely to route UEs.

Motivated by these gaps, this work develops a unified framework comprising LinUCB-based UE association and REINFORCE-based drone mobility optimisation, evaluated under a comprehensive 6G propagation model incorporating sub-THz molecular absorption, 3D beamforming across four BS tiers, and a physically grounded battery lifecycle model for aerial base stations.

III. Experimental Setup

A. 6G Grid Conditions Considered

To evaluate learning-based handover strategies in realistic sixth-generation (6G) conditions, we construct a multi-layer heterogeneous cellular grid that models the spatial, spectral, and propagation characteristics expected in future networks. The simulation includes macro, micro, pico, and drone base stations, each operating on sub-6, mmWave, and sub-THz carrier frequencies. This layered architecture enables both wide-area and highly localised coverage, reflecting the density, beam dependence, and ultra-high-frequency operation anticipated in 6G deployments. The addition of a mobile drone tier further allows the network to respond dynamically to spatial variations in user demand and coverage gaps that static infrastructure cannot address.

*1) **Macro cells**:* constitute the highest-power layer and provide the widest coverage region. They ensure global connectivity across the grid, especially in areas lacking dense deployment. Macro cells are placed in a circular ring at a fixed radius from the grid centre to provide uniform wide-area coverage. Due to their large coverage footprint, macro cells deliver lower beam precision and experience slower variations in link quality but are essential for mobility continuity.

*2) **Micro cells**:* occupy the intermediate tier, operating at moderate transmit power and covering smaller regions within the macro layer. They relieve macro-cell load and create smoother handover transitions by offering higher capacity in moderately dense areas. The reduced coverage radius results in more distance-sensitive path loss and more frequent beam fluctuations than in macro-tier links.

*3) **Pico cells**:* form the densest ground layer and are deployed in hotspots requiring extremely high capacity. Their low transmit power and short-range operation greatly amplify the impact of blockage and small-scale fading, especially at mmWave and sub-THz frequencies. Pico cells naturally induce frequent handovers because users traverse their coverage areas rapidly. Their dense placement makes them ideal for evaluating mobility performance in beam-based 6G environments.

*4) **Drone Base Station**:* constitute the novel fourth tier introduced in this work. Unlike the static ground tiers above, drone BSs are aerial and mobile, operating at altitudes between 30 m and 150 m and repositioning each simulation step under the control of individual Policy Gradient agents. Six drone BSs are deployed within the inner 80% of the simulation area with randomised initial positions and altitudes. Their mobility allows them to fill coverage gaps, serve UEs that lack a strong ground alternative, and redistribute load dynamically. Drone BSs are battery-constrained: when their state-of-charge drops to the critical threshold, they return to their ground charging station before relaunching, temporarily withdrawing from the network.

Together, the four tiers create a hierarchical HetNet in which coverage responsibility is distributed across wide-area, mid-range, ultra-dense, and mobile aerial access points. This arrangement generates highly heterogeneous link conditions and realistic handover triggers, forming a comprehensive testbed for the joint evaluation of LinUCB-based UE association and Policy Gradient drone mobility.

B. Base Station Parameters

Each BS tier has specific technical features to reflect its role in the heterogeneous architecture. Macro BSs provide wide-area coverage, micro BSs enhance capacity in medium-density areas, pico BSs target high-density hotspots, and drone BSs provide mobile aerial coverage with battery-aware operation.

C. Network Topology

The simulation is conducted over a 3000×3000 m^2 area, consisting of a four-tier heterogeneous network with 4 macro, 8 micro, 12 pico, and 6 drone base stations, totalling 30 BSs. The network serves 300 user equipments (UEs) positioned at a height of 1.5 m with random initial velocities in the range $[-2, 2]$ m/step.

TABLE I
NETWORK TOPOLOGY PARAMETERS

Parameter	Value
Simulation area	3000×3000 m^2
Macro BSs	4
Micro BSs	8
Pico BSs	12
Drone BSs	6
Total BSs	30
Number of UEs	300
UE height	1.5 m
Simulation steps	200
Step duration	1 s

TABLE II
BASE STATION TECHNICAL FEATURES BY TIER

BS Tier	Height (m)	Antennas	Max Beams	BF Gain (dB)	Tx Power (dBm)
Macro	30	128	256	25	43
Micro	15	256	128	30	35
Pico	8	512	64	35	28
Drone	30–150	64	32	20	38
UE	1.5	4/8/16	–	–	–

D. Drone and Battery Parameters

The drone tier introduces a battery lifecycle that governs each drone's operational availability. Table III lists the battery and mobility parameters used in all experiments. Each drone is initialised with a randomised state-of-charge between 60% and 100% to simulate staggered fleet deployment. When the state-of-charge falls to the critical threshold, the drone enters the Return-To-Base (RTB) state and disconnects all connected UEs, which are immediately reassociated by LinUCB. Once the drone recharges to the resume threshold at its ground charging station, it relaunches and re-enters the active serving pool.

TABLE III
DRONE AND BATTERY SYSTEM PARAMETERS

Parameter	Value
Number of drone BSs	6
Altitude range	30–150 m
Maximum speed	15 m/step
Battery capacity	500 Wh
Critical SoC threshold	10%
Resume SoC threshold	90%
Hover power	800 W
Horizontal movement coefficient	3 W/(m/step)
Vertical movement coefficient	5 W/(m/step)
RF electronics overhead	15 W
Charging rate	1200 W
Initial SoC (randomised)	60–100%
PG update interval	Every 10 steps

IV. CMAB PROBLEM FORMULATION

The problem is formulated as a Contextual Multi-Armed Bandit (CMAB) instead of a deterministic or static optimisation framework because the wireless environment is highly dynamic, uncertain, and time-varying. The achievable throughput, interference, and handover impact are not known a priori and change continuously due to user mobility, fading, varying network load, and the repositioning of drone base stations. CMAB enables online learning directly from real-time observations while exploiting contextual information, without requiring an accurate analytical model of the environment. This makes the framework well suited for adaptive user association and handover management in dynamic 6G heterogeneous networks with both static and mobile base stations.

At every time step, each user equipment (UE) observes the wireless environment, selects one serving base station (BS) from the pool of currently active base stations, and receives a scalar reward that reflects both throughput performance and mobility cost. Drone BSs that are in the RTB or Charging state are excluded from the active arm set for that step, and any UE previously connected to a drone that has gone offline is forced to reselect from the remaining active BSs.

A. Arms

Let the set of candidate BSs available to a UE at time t be

$$\mathcal{A}_t = \{1, 2, \ldots, N_t\}, \tag{2}$$

where each arm $a \in \mathcal{A}_t$ represents a possible serving BS. Selecting an arm corresponds to associating the UE with that BS for the current time step. In the extended four-tier simulation, $N_t \leq 30$ at any step, since the effective arm count decreases whenever one or more drone BSs are offline for charging or returning to base. When a drone BS returns to the active state after recharging, the LinUCB agent dynamically re-admits it as a valid arm, restoring its previously learned parameter estimates without resetting the other arms.

B. Context

For each UE–BS pair (t, a), a context vector

$$\mathbf{x}_{t,a} \in \mathbb{R}^d \tag{3}$$

is observed, containing real-time radio and network state information. The context vector is ten-dimensional ($d = 10$) and includes:

- normalised instantaneous SINR,
- normalised 3D distance to BS,
- normalised azimuth and elevation angles,
- normalised carrier frequency,
- normalised transmit power,
- normalised bandwidth,
- normalised antenna count,
- normalised BS load, and
- a binary sub-THz band indicator.

The context captures the time-varying nature of the wireless environment and guides the learning process. For drone BSs, the context vector additionally reflects altitude-dependent path loss corrections and the narrower beamwidth associated with aerial links, so the agent implicitly learns that the value of a drone arm is sensitive to its current position relative to the UE.

C. Reward Variables

The reward at each time step is defined using the following observable variables:

$$\text{Rewards: } \{\text{Throughput, Handover (Boolean), Previous Throughput, SINR}\}.$$

The base reward is computed from the current achieved throughput as

$$R_{\text{base}} = \frac{\text{Throughput}}{1000}. \tag{4}$$

A fixed penalty is assigned for any handover event that does not yield a meaningful throughput improvement:

$$H_o = 0.1. \tag{5}$$

A bonus is awarded when the selected BS provides a high-quality signal:

$$B_{\text{SINR}} = 0.05 \cdot \mathbb{1}[\text{SINR} > 20\,\text{dB}]. \tag{6}$$

The relative improvement in throughput is defined as

$$\Delta = \text{Throughput} - \text{Previous Throughput}. \tag{7}$$

D. Final Reward Function

The final reward used by the CMAB agent is constructed as

$$R_t = \begin{cases} R_{\text{base}} - H_o + B_{\text{SINR}}, & \text{if handover occurs and } \Delta < 25\,\text{Mbps}, \\ R_{\text{base}} + B_{\text{SINR}}, & \text{otherwise.} \end{cases} \tag{8}$$

This formulation explicitly penalises unnecessary handovers that do not provide a sufficient throughput improvement, thereby discouraging frequent or low-gain handovers while still promoting high data-rate and high signal quality links. The SINR bonus further incentivises the agent to prefer BSs that provide stable, high-quality connections — a property of particular importance when choosing between a static ground BS and a mobile drone BS whose position may be temporarily suboptimal. This reward structure remains identical across all three drone mobility algorithms evaluated in this work, ensuring that observed performance differences are attributable solely to the choice of policy gradient method rather than any change in the association objective.

E. Learning Objective

The CMAB agent aims to learn an optimal association policy that maximises the expected long-term cumulative reward:

$$\max \mathbb{E}\left[\sum_{t=1}^{T} R_t\right]. \tag{9}$$

This objective jointly optimises:

- long-term throughput,
- mobility stability,
- handover efficiency,
- reduction of unnecessary handovers, and
- preference for high signal quality links.

F. Drone Policy Gradient Formulation

In addition to the CMAB problem for UE association, the repositioning of each drone BS is formulated as a separate sequential decision-making problem. Each active drone observes a 13-dimensional state vector $\mathbf{s}_t$ encoding its own position, the centroid of its connected UEs, load, coverage score, distances to neighbouring BSs, recent handover rate, outage rate, battery state-of-charge, and distance to its charging station. The drone then selects a continuous three-dimensional movement action $\mathbf{a}_t = (\Delta x, \Delta y, \Delta z)$ from a Gaussian policy $\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)$.

The drone reward r_t integrates coverage quality, handover penalty, gap-fill bonus, energy efficiency, low-battery urgency, and boundary avoidance into a single scalar signal that reflects both network performance and operational constraints. This reward is state-dependent: drones in the CHARGING state receive a charge-progress bonus minus a coverage-loss penalty; drones in the RTB state receive a flat per-step penalty plus a progress bonus; and active drones receive the full six-component reward. This reward structure is held constant across all three policy gradient methods so that their performance can be compared fairly.

The drone policy objective is to maximise the expected discounted cumulative reward:

$$\max_{\theta} \mathbb{E}_{\pi_\theta}\left[\sum_{t=1}^{T} \gamma^t r_t\right], \quad \gamma = 0.97. \tag{10}$$

Three policy gradient algorithms are evaluated as solutions to this optimisation problem, each differing in how it estimates the advantage and utilises collected experience.

*1) **REINFORCE with Baseline**:* collects experience over a fixed horizon of ten simulation steps, computes full Monte Carlo discounted returns, and subtracts a learned state-value baseline to reduce gradient variance. It is entirely on-policy and requires no replay buffer, making it simple to implement but sensitive to the high variance of Monte Carlo returns, particularly when battery state transitions abruptly change the reward structure mid-trajectory.

*2) **Advantage Actor-Critic (A2C)**:* replaces the Monte Carlo return with a one-step bootstrapped TD estimate, computing the advantage as the difference between the observed reward plus the discounted next-state value and the current-state value. This localises the advantage estimate to a single transition, making it naturally robust to the hard state changes introduced by the battery finite state machine, since a transition from ACTIVE to RTB affects only the single step at which it occurs rather than corrupting an entire trajectory of returns.

*3) **Soft Actor-Critic (SAC)**:* additionally augments the reward with an entropy bonus that encourages the drone to maintain a diverse movement policy throughout training, preventing premature convergence to suboptimal positioning strategies. SAC is off-policy: it stores all past transitions in a replay buffer and samples random minibatches for each update, allowing it to reuse the full history of the 200-step simulation rather than discarding experience after each on-policy update. Twin critic networks with soft target updates further stabilise value estimation. The temperature parameter α, which controls the trade-off between reward maximisation and entropy, is tuned automatically during training.

The two optimisation problems — LinUCB for UE association and the chosen Policy Gradient method for drone mobility — run simultaneously and interact throughout the simulation: drone repositioning changes the SINR landscape observed by LinUCB, and LinUCB's association decisions change the load and coverage score observed by each drone's policy network. This joint optimisation is not explicitly coordinated but emerges naturally from the shared environment, and its behaviour differs across the three policy gradient methods as each learns a distinct repositioning strategy in response to the evolving network state.

V. Problem Solution Using Bandit-Based and Policy Gradient Learning

The CMAB-formulated user association and handover problem is solved using LinUCB as a context-aware solution that exploits real-time network information. Drone mobility is solved simultaneously by independent policy gradient agents, with three algorithms evaluated: REINFORCE with baseline, Advantage Actor-Critic (A2C), and Soft Actor-Critic (SAC). A time-to-trigger mechanism is layered on top of LinUCB to suppress impulsive handovers caused by momentary signal fluctuations. The LinUCB component remains identical across all three configurations, ensuring that observed performance differences are attributable solely to the drone mobility algorithm.

A. LinUCB as a Context-Aware Solution

To incorporate instantaneous network conditions into the learning process, a context-aware LinUCB framework is adopted [**?**]. In this approach, each UE observes a feature vector $\mathbf{x}_{t,a}$ for every UE–BS pair, which captures real-time radio and network conditions including SINR, distance, beam alignment, frequency band, and BS load.

The expected reward is assumed to follow a linear model:

$$\mathbb{E}[R_t \mid \mathbf{x}_{t,a}] = \mathbf{x}_{t,a}^\top \boldsymbol{\theta}_a, \tag{11}$$

where $\boldsymbol{\theta}_a$ is an unknown parameter vector associated with BS a. For each UE u, separate ridge regression matrices $\mathbf{A}_{u,a} \in \mathbb{R}^{d\times d}$ (initialised as $\mathbf{I}_d$) and vectors $\mathbf{b}_{u,a} \in \mathbb{R}^d$ (initialised as $\mathbf{0}$) are maintained per arm. At each time step, the UE selects the BS that maximises the following upper confidence bound:

$$a^* = \arg\max_{a\in\mathcal{A}_t} \left[\mathbf{x}_{t,a}^\top \hat{\boldsymbol{\theta}}_a + \alpha\sqrt{\mathbf{x}_{t,a}^\top \mathbf{A}_a^{-1}\,\mathbf{x}_{t,a}}\right], \tag{12}$$

where a handover penalty of 0.35 is subtracted from the UCB score of any arm that is not the current serving cell. This hysteresis term discourages unnecessary switching when the throughput gain does not justify the handover cost.

After observing the reward R_t, model parameters are updated as:

$$\mathbf{A}_{a^*} \leftarrow \mathbf{A}_{a^*} + \mathbf{x}_{t,a^*}\mathbf{x}_{t,a^*}^\top, \tag{13}$$

$$\mathbf{b}_{a^*} \leftarrow \mathbf{b}_{a^*} + R_t\,\mathbf{x}_{t,a^*}, \tag{14}$$

$$\hat{\boldsymbol{\theta}}_{a^*} \leftarrow \mathbf{A}_{a^*}^{-1}\mathbf{b}_{a^*}. \tag{15}$$

Algorithm 1 Context-Aware LinUCB-Based UE Association with Drone Awareness

```
Initialise for each arm a: A_a = I_d, b_a = 0_d
for t = 1 to T do
    Determine active arm set A_t (exclude offline drones)
    for each arm a ∈ A_t do
        Observe context x_{t,a}
        Compute θ̂_a = A_a^{-1} b_a
        Compute UCB score: p_a(t) = x_{t,a}^T θ̂_a + α sqrt(x_{t,a}^T A_a^{-1} x_{t,a})
        if a ≠ current serving BS then
            p_a(t) ← p_a(t) − 0.35 {Handover penalty}
        end if
    end for
    Select a* = arg max_{a∈A_t} p_a(t)
    if a* ≠ current BS and TTT counter < τ then
        Increment TTT counter; retain current BS
    else
        Execute handover to a*; reset TTT counter
    end if
    Observe reward R_t
    Update: A_{a*} ← A_{a*} + x_{t,a*} x_{t,a*}^T,  b_{a*} ← b_{a*} + R_t x_{t,a*}
end for=0
```

B. Time-to-Trigger Mechanism

A time-to-trigger (TTT) filter is applied on top of LinUCB to prevent ping-pong handovers caused by transient signal fluctuations. A handover from the current BS to a candidate BS a^* is only executed if a^* has been selected by LinUCB for $\tau = 5$ consecutive steps. If a different arm is selected before the counter reaches τ, the counter resets to one for the new candidate. This mechanism reflects standard 3GPP handover procedure while remaining fully compatible with the online learning update, since the LinUCB model is updated at every step regardless of whether a handover is executed.

C. REINFORCE with Baseline for Drone Mobility

Each drone BS is controlled by an independent REINFORCE agent with a learned value baseline [**?**]. The policy is parameterised by a two-layer MLP with a policy head that outputs the mean of a Gaussian action distribution over three-dimensional movements $(\Delta x, \Delta y, \Delta z)$, and a separate value head that estimates the expected return from the current state.

The policy selects actions by sampling from the Gaussian policy:

$$\mathbf{a}_t \sim \mathcal{N}\big(\boldsymbol{\mu}_\theta(\mathbf{s}_t),\ \mathrm{diag}(\boldsymbol{\sigma}^2)\big)\,, \tag{16}$$

where $\boldsymbol{\mu}_\theta(\mathbf{s}_t) = \tanh(\mathbf{W}_2^\mu \mathbf{h} + \mathbf{b}_2^\mu) \cdot v_{\max}$ and $\log \boldsymbol{\sigma}$ is a learnable parameter vector. Actions are clipped to $[-v_{\max}, v_{\max}]$ in each dimension.

Every $\Delta_{\text{update}} = 10$ steps, the agent performs a REINFORCE update using the collected episode trajectory. Discounted returns are computed and normalised, and the policy weights are updated by gradient ascent on the advantage-weighted log-probability:

$$\nabla_\theta J(\theta) = \frac{1}{T} \sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t) \cdot \hat{A}_t, \tag{17}$$

where $\hat{A}_t = G_t - V_\phi(\mathbf{s}_t)$ is the advantage estimate. The value head is updated simultaneously by minimising the mean squared error between $V_\phi(\mathbf{s}_t)$ and G_t, reducing variance in the gradient estimates over time.

Algorithm 2 Battery-Aware REINFORCE for Drone BS Mobility

Initialise policy weights θ, value weights ϕ, episode buffer $\mathcal{B} = \emptyset$
for $t = 1$ to T **do**
  Determine drone state (ACTIVE, RTB, or CHARGING)
  if drone state $=$ ACTIVE **then**
    Observe state $\mathbf{s}_t$; sample action $\mathbf{a}_t \sim \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{s}_t), \mathrm{diag}(\boldsymbol{\sigma}^2))$
    if SoC $\leq \beta_{\text{crit}}$ **then**
      Override action $\rightarrow$ transition to RTB
    else
      Apply movement; consume battery energy E_{step}
    end if
  else if drone state $=$ RTB **then**
    Fly toward charging station at $1.5\times$ max speed
    On arrival $\rightarrow$ transition to CHARGING
  else if drone state $=$ CHARGING **then**
    Recharge at 1200 W; if SoC $\geq \beta_{\text{resume}} \rightarrow$ ACTIVE
  end if
  Compute reward r_t; store $(\mathbf{s}_t, \mathbf{a}_t, r_t)$ in $\mathcal{B}$
  if $t \bmod \Delta_{\text{update}} = 0$ **then**
    Compute discounted returns G_t and normalise
    Update θ via $\frac{1}{|\mathcal{B}|}\sum \log \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)(G_t - V_\phi(\mathbf{s}_t))$
    Update ϕ by minimising $\frac{1}{|\mathcal{B}|}\sum (G_t - V_\phi(\mathbf{s}_t))^2$
    Clear $\mathcal{B} \leftarrow \emptyset$
  end if
end for=0

D. Advantage Actor-Critic (A2C) for Drone Mobility

The A2C variant replaces the Monte Carlo return of REINFORCE with a one-step bootstrapped TD advantage, computed at every simulation step as:

$$\hat{A}_t = r_t + \gamma V_\phi(\mathbf{s}_{t+1}) - V_\phi(\mathbf{s}_t). \tag{18}$$

This localises the advantage estimate to a single transition, making A2C naturally robust to the hard state changes introduced by the battery finite state machine. When a drone transitions from ACTIVE to RTB, the reward structure changes abruptly; in REINFORCE this corrupts the entire Monte Carlo return for that trajectory, whereas in A2C the disruption is confined to the single step at which the transition occurs. The policy and value network architectures are identical to those used in REINFORCE, and the same Gaussian action sampling and clipping are applied. The policy is updated at every step rather than every ten steps, with no episode buffer required. An entropy bonus $\beta \cdot \mathcal{H}(\pi)$ is added to the actor loss to maintain exploration throughout training.

Algorithm 3 Battery-Aware A2C for Drone BS Mobility

Initialise policy weights θ, value weights ϕ
for $t = 1$ to T **do**
  Determine drone state; handle RTB and CHARGING as in Algorithm 2
  if drone state $=$ ACTIVE **then**
    Observe $\mathbf{s}_t$; sample $\mathbf{a}_t \sim \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{s}_t), \mathrm{diag}(\boldsymbol{\sigma}^2))$
    Apply movement; consume battery energy E_{step}
    Observe reward r_t and next state $\mathbf{s}_{t+1}$
    Compute TD advantage: $\hat{A}_t = r_t + \gamma V_\phi(\mathbf{s}_{t+1}) - V_\phi(\mathbf{s}_t)$
    Update θ via $\nabla_\theta \log \pi_\theta(\mathbf{a}_t|\mathbf{s}_t) \cdot \hat{A}_t + \beta \nabla_\theta \mathcal{H}(\pi_\theta(\cdot|\mathbf{s}_t))$
    Update ϕ by minimising $(r_t + \gamma V_\phi(\mathbf{s}_{t+1}) - V_\phi(\mathbf{s}_t))^2$
  end if
end for=0

E. Soft Actor-Critic (SAC) for Drone Mobility

The SAC variant extends the policy gradient framework by incorporating three additional mechanisms that address key limitations of on-policy methods in the short-horizon, non-stationary setting of this simulation [**?**]. First, an *entropy bonus* augments the reward at every step, encouraging the drone to maintain a diverse movement policy and preventing premature convergence to a fixed hovering strategy:

$$r_t^{\text{soft}} = r_t + \alpha \cdot \mathcal{H}(\pi_\theta(\cdot \mid \mathbf{s}_t)), \tag{19}$$

where α is a temperature parameter tuned automatically during training to maintain a target entropy level of $-|\mathcal{A}|$. Second, an *off-policy replay buffer* stores all past transitions (s_t, a_t, r_t, s_{t+1}) and supplies random minibatches for each gradient update, allowing the agent to reuse the full history of the 200-step simulation rather than discarding experience after each on-policy update. Third, *twin critic networks* Q_1 and Q_2 with soft target updates are used to estimate the state-action value, with the minimum of the two critics taken as the update target to reduce overestimation bias:

$$y_t = r_t + \gamma \left(\min_{i=1,2} Q_i(\mathbf{s}_{t+1}, \mathbf{a}_{t+1}) - \alpha \log \pi_\theta(\mathbf{a}_{t+1} \mid \mathbf{s}_{t+1}) \right). \tag{20}$$

Target networks are updated via soft polyak averaging $\theta_{\text{target}} \leftarrow \tau\theta + (1-\tau)\theta_{\text{target}}$ with $\tau = 0.005$, providing stable learning targets throughout training. The actor network architecture remains identical to that used in REINFORCE and A2C, with actions produced via the reparameterisation trick $\mathbf{a}_t = \tanh(\boldsymbol{\mu}_\theta + \boldsymbol{\sigma}\boldsymbol{\varepsilon}) \cdot v_{\max}$, $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, to allow gradient flow through the sampling operation.

F. Joint Operation of LinUCB and Policy Gradient

At each simulation step, the two learning systems operate in the following sequence regardless of which policy gradient algorithm is active. First, all UEs execute LinUCB arm selection from the currently active BS pool, with the TTT filter applied to candidate handovers. Rewards are observed and LinUCB models updated. Second, drone movement actions are selected by each drone's policy gradient agent based on the post-association network state — so the drone reward correctly reflects the coverage quality after LinUCB has made its UE association decisions. Finally, battery state machines are updated and drone state transitions are applied. This ordering ensures that the drone reward signal is meaningful: it reflects actual UE connections and SINR values rather than a pre-association estimate, and LinUCB always operates on the current drone positions rather than stale coordinates.

The dynamic interplay between the two agents naturally produces cooperative behaviour without explicit coordination: LinUCB steers UEs toward whichever BS currently offers the best signal (including drones), while each drone's policy learns to reposition toward under-served UEs and coverage gaps where LinUCB is most likely to route traffic to it. This interaction manifests differently across the three policy gradient methods — REINFORCE converges more slowly due to high-variance Monte Carlo updates; A2C adapts more quickly to battery state transitions through per-step TD updates; and

Algorithm 4 Battery-Aware SAC for Drone BS Mobility

Initialise actor θ, critics ϕ_1, ϕ_2, targets $\bar{\phi}_1, \bar{\phi}_2$, replay buffer $\mathcal{D}$, $\log \alpha$
for $t = 1$ to T **do**
    Determine drone state; handle RTB and CHARGING as in Algorithm 2
    if drone state $=$ ACTIVE **then**
        Observe $\mathbf{s}_t$; sample $\mathbf{a}_t$ via reparameterisation
        Apply movement; observe r_t, $\mathbf{s}_{t+1}$; store $(\mathbf{s}_t, \mathbf{a}_t, r_t, \mathbf{s}_{t+1})$ in $\mathcal{D}$
        if $|\mathcal{D}| \geq$ min buffer size **then**
            Sample minibatch from $\mathcal{D}$
            Compute soft TD target $y_t = r_t + \gamma(\min_i Q_{\bar{\phi}_i}(\mathbf{s}_{t+1}, \mathbf{a}_{t+1}) - \alpha \log \pi_\theta)$
            Update ϕ_1, ϕ_2 by minimising $(Q_{\phi_i} - y_t)^2$
            Update θ by maximising $\mathbb{E}[\min_i Q_{\phi_i}(\mathbf{s}, \mathbf{a}_\pi) - \alpha \log \pi_\theta(\mathbf{a}_\pi|\mathbf{s})]$
            Update $\log \alpha$ to minimise $-\alpha(\log \pi_\theta + \mathcal{H}_{\text{target}})$
            Soft-update targets: $\bar{\phi}_i \leftarrow \tau\phi_i + (1-\tau)\bar{\phi}_i$
        end if
    end if
end for=0

SAC maintains broader exploration throughout the simulation due to its entropy regularisation and replay buffer, potentially discovering positioning strategies that on-policy methods miss within the limited 200-step horizon.